City Opportunities: Data, Evaluation and Model Extension

Evidence-based bounds support a conservative city recommendation while retaining missing observations. In the sixteen-city comparison, verified supplementation narrows average score intervals by 13.97 %. Zurich minimizes maximum regret under the baseline weights, while all sixteen cities remain possible champions.

4 Data collection and comparability

4.1 A traceable city–indicator matrix

We compare $M = 16$ cities on $N = 10$ indicators covering safety, environment, household costs, education and everyday resources. The initial matrix is the team-supplied city table [1]. We retain its existing values, populations, reference years and source flags as the agreed input, then supplement only empty cells. Some supplied values lack original source URLs, so their provenance remains the team table; official supplements do not retrospectively validate every supplied measurement. Table 1 states what each variable measures.

Table 1: Retained indicators, units and reference anchors. Supplied values are retained; supplements have explicit source scope.

Indicator	Measurement and interpretation	Anchor
Homicide	Deaths per 100,000; cost; supplied city proxy	2
$PM_{2.5}$	$\mu g/m^3$; cost; supplied concentration proxy	15
Living costs	Supplied living-cost index; cost	100
Housing	Supplied price-to-income ratio; cost	10
School resources	Pupils per teacher or explicitly scoped FTE; cost	15
Higher education	Institutions per 100,000; benefit; supply proxy	1
Transport stops	Unique stops per 100,000; benefit; partial inventories	50
Travel	Duration in minutes; $\exp(-t/30)$; commute or stage proxy	30 min
Broadband	Supplied Mbps; benefit; city/general-use proxy	100
Swimming pools	Venues per 100,000; benefit; supply proxy	1

The nominal reference year is 2025, but the evidence is mixed-vintage. Every supplement retains its actual year rather than being relabelled as 2025. Population denominators remain

those of the supplied table. Thus a 2026 facility snapshot divided by a supplied 2025 population is a conditional density, not an observed same-year census. We do not infer that facilities are unchanged merely because they may change slowly. Historical inventory assumptions are evaluated through withdrawal scenarios.

We use official statistical tables, operator inventories and municipal geodata to fill missing fields. For each admitted supplement, the archive records the provider, URL, year, geographic and population scope, units, extraction rule and file hash. Original PDFs and API responses are retained separately from derived tables. Search summaries locate evidence; numerical admission requires the underlying record or an explicitly identified provider capture. Values are converted to compatible units before utility calculation: for example, the supplied higher-education density per million is divided by ten to obtain the per-100,000 scale. Mode share is not converted into physical stop density.

4.2 Collection problems and their resolution

A station is not a platform, and a network is not a city. Oslo's municipal GIS represents platforms. Selecting records within the city boundary and consolidating names gives 89 metro stations from 186 platforms. Zurich's official 2025 VBZ report separately lists 446 stops on city territory and 235 in the surrounding area; only 446 enter the city inventory [5, 6]. A November 2026 snapshot of Helsinki's operator similarly distinguishes 19 metro stations in Helsinki from 11 in Espoo [7]. These are mode or operator subsets, so they supply lower bounds rather than complete all-operator totals. For a verified facility count c and fixed population P, density is $d = 10^5 c/P$; utility is bounded by $[d/(a+d), 1]$.

A list of venues can contain closed facilities. Stockholm's current municipal swimming directory lists fourteen named venues, one of which is explicitly closed for renovation. The thirteen retained venues agree with the current overview; the temporary year-round replacement is counted once [8]. We count venues, not individual basins. This municipal subset gives a supply lower bound and leaves other providers unknown. Its access date is November 2026; a page-footer date is not used to reconstruct a 2025 inventory. Paris's 244 intra-muros metro stations come from a March 2022 atlas using RATP 2021 and Apur 2022 sources, not a 2025 observation [9]. A separate withdrawal restores this historical supplement to unknown.

Pupils and teachers must refer to the same schools and stage. Prague's 2024/25 tables report 80,073 secondary-school pupils and 6,877.1 teacher full-time equivalents (FTE), including lower-gymnasium years. Subtracting the matching 8,748 pupils and 737.8 FTE yields 71,325 pupils and a rounding envelope of $[6139.2, 6139.4]$ FTE [10]. The Prague measure includes all study modes, including adult and non-daytime cohorts. Stockholm's published city-school ratio covers upper-secondary years 1–3 and all provider ownership types, measured on 15 October [11]. Its API value 16.42091405 is conservatively represented by $[16.35, 16.45]$, an intentional precision convention rather than a claimed measurement-error confidence interval. In contrast, Paris's 72,476 lycée pupils cannot be divided by its 15,940 second-degree teachers because the teacher total includes collèges [12]; that missing ratio remains unknown. FTE resource measures and school-location populations are distinguished from teacher headcounts and resident high-school outcomes.

Unavailable records and unobserved travellers are retained honestly. An initial Singapore census PDF endpoint returned 404; the official Parliament mirror and official data API supplied the report and matching table [13]. Duration bins are converted using utility at

their endpoints, without bin midpoints or an artificial maximum duration for the open tail. The stage-specific total enrolment estimates exceed the travel-table estimates by estimated uncovered student totals of 1,811 for secondary, 989 for post-secondary non-tertiary and 1,635 for polytechnic. These are weighted survey estimates rather than complete headcounts. The uncovered mass retains utility $[0, 1]$; unknown stage proportions are handled by the interval convex hull. The resulting $[0.21048, 0.61884]$ is a 2020 school-stage proxy, not an identified 2025 interval for ages 14–18. When Dublin raw stop downloads returned 403, no count was inferred from the catalogue. A later official map supported a separately documented subset of 19 existing city-centre tram stops [14]. A timed-out Warsaw school-travel download was discarded and the field left unknown. Failed retrieval does not create a numerical observation.

The current 160 cells comprise 112 point values, 27 partial intervals and 21 fully missing intervals. The original table had 112 points, 12 partial intervals and 36 fully missing cells. The fifteen supplements therefore improve information without replacing the team's existing measurements. Facility supply, general commute proxies and school resources remain distinct from adolescent access, school travel and learning quality.

5 Establishing the baseline evaluation model

5.1 Determining indicator weights

We retain eight main indicators at weight 0.1125 each and two auxiliary facility indicators, transport stops and pools, at 0.05 each, so $\sum_j w_j = 1$. These weights express a transparent modelling choice. They are not estimated from a student survey. Equal weights provide a separate reference. TOPSIS is the primary score; VIKOR provides a focused method comparison. AHP represents explicit preference scenarios [2], and Monte Carlo measures sensitivity to declared distributions. These extensions retain the same criteria rather than introduce unrelated ranking models.

For preference multipliers $m_j > 0$, we use

$$w'_j = \frac{m_j w_j}{\sum_k w_k m_k}, \qquad A_{jk} = \frac{w'_j}{w'_k}. \tag{1}$$

The principal eigenvector of A recovers w'. With $\lambda_{\max}$ its largest eigenvalue, $CI = (\lambda_{\max} - N)/(N-1)$ and $CR = CI/RI_N$. Because this matrix is built from exact ratios, $\lambda_{\max} = N$ and $CR = 0$ up to numerical precision. This verifies internal consistency; it does not establish that these are students' actual preferences.

5.2 TOPSIS scoring calculation

We adapt TOPSIS [3] using fixed reference anchors to preserve a common scoring scale across evidence, preference and future scenarios. For benefit and cost measurements respectively,

$$u^+(x) = \frac{x}{a+x}, \qquad u^-(x) = \frac{a}{a+x}. \tag{2}$$

Each anchor in Table 1 is a declared reference scale, not an estimated optimum. Travel uses $q(t) = e^{-t/30}$ for duration t in minutes. When only mean duration $\bar{t}$ is known, Jensen's inequality gives $\mathbb{E}[q(t)] \in [e^{-\bar{t}/30}, 1]$. Fully missing utilities remain $[0, 1]$.

For utility vector u_i, weighted distances from fixed worst and best points are

$$D_i^- = \sqrt{\sum_j w_j u_{ij}^2}, \qquad D_i^+ = \sqrt{\sum_j w_j (1-u_{ij})^2}, \qquad S_i = 100 \frac{D_i^-}{D_i^- + D_i^+}. \tag{3}$$

Weights enter the squared distances once. Since S_i increases with every utility, endpoints ℓ_i, h_i give $L_i = S(\ell_i)$ and $U_i = S(h_i)$. Under the retained rectangular uncertainty set, define

$$R_i = \max\{0, \max_{k \neq i} U_k - L_i\}, \qquad i^* \in \arg\min_i R_i, \tag{4}$$

$$\mathcal{C} = \{i : U_i \geq \max_{k \neq i} L_k\}. \tag{5}$$

Maximum regret R_i measures how far the selected city could trail another city. Set $\mathcal{C}$ retains every city that could still be first. The maximum excludes the selected city itself. These conditional data bounds are not sampling confidence intervals, and a conservative recommendation does not identify a uniquely best city.

5.3 Scoring results and interpretation

Zurich minimizes maximum regret, followed by Singapore and Oslo (Table 2). Zurich's score lower bound is below those of Singapore and Oslo; its own upper bound is excluded from its competitor set. Its regret is $70.90353 - 53.53548 = 17.36805$, whereas Oslo's is $72.18379 - 54.29594 = 17.88785$. All sixteen cities remain possible champions. The decision order expresses a conservative recommendation under the stated criteria and intervals.

Table 2: Conservative baseline decision ranks with unchanged weights.

Rank	City	Score interval	Maximum regret
1	Zurich	53.54–72.18	17.37
2	Singapore	54.33–63.53	17.85
3	Oslo	54.30–63.90	17.89
4	Stockholm	52.12–64.54	20.06
5	Prague	51.62–62.78	20.56
6	Dublin	51.13–70.90	21.05
7	Berlin	50.49–61.54	21.69
8	Helsinki	49.41–69.84	22.78
9	Paris	49.11–69.04	23.08
10	Madrid	48.63–68.22	23.56
11	Warsaw	47.50–63.99	24.69
12	Vienna	47.19–61.03	25.00
13	Milan	45.63–65.99	26.56
14	London	45.17–66.12	27.01
15	Tokyo	43.81–61.12	28.38
16	Amsterdam	42.29–60.43	29.90

Holding weights and formulas fixed, the fifteen admitted supplements narrow the average fixed-weight score interval from 18.81077 to 16.18302, a 13.97 % reduction. The most recent five additions alone narrow it by 4.44%, moving Stockholm from thirteenth to fourth.

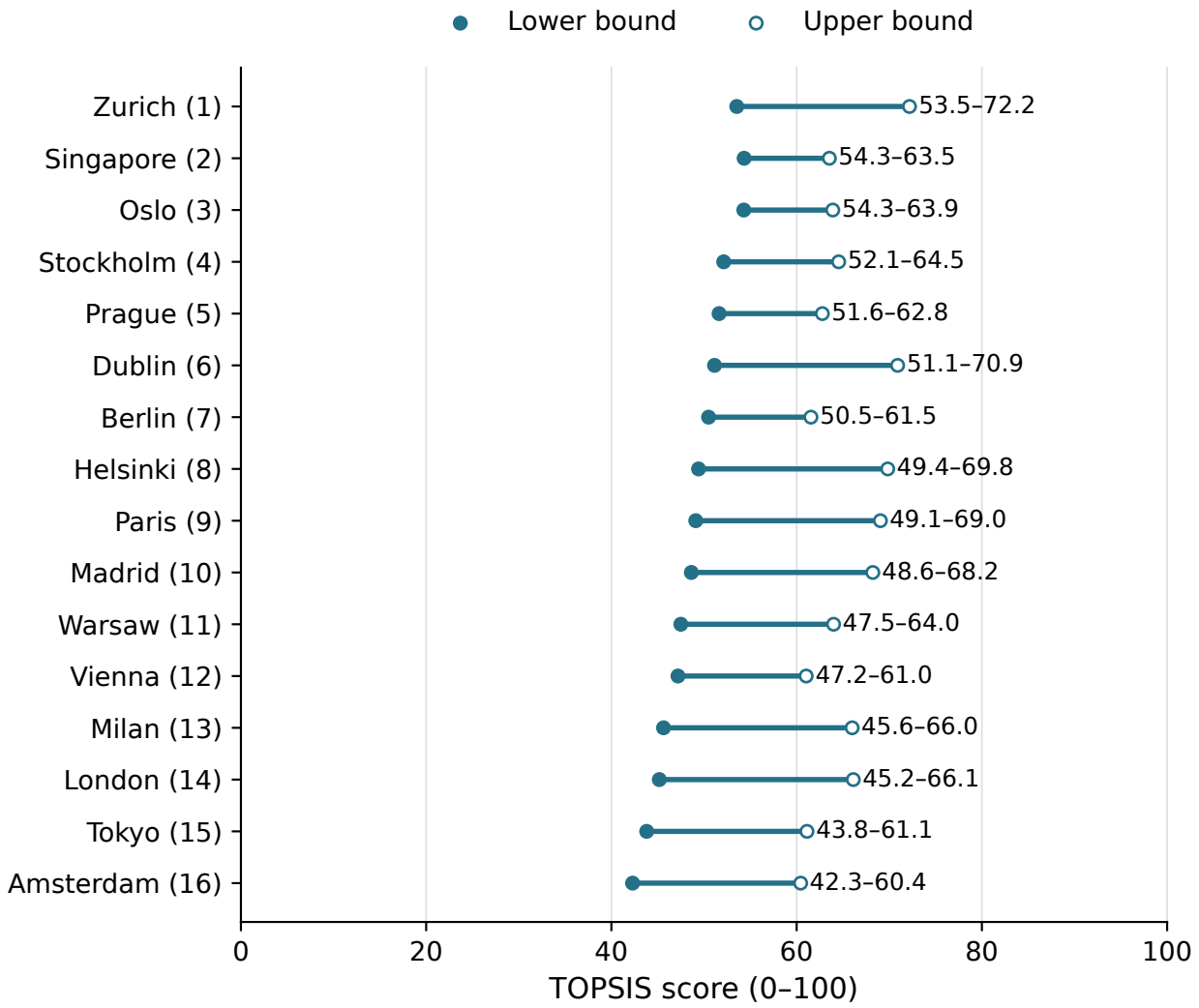


Figure 1: Current score bounds and conservative decision order for the sixteen cities. Intervals retain missing information; they are not confidence intervals. Source: frozen city matrix and admitted supplements.

Withdrawal of the two recent historical transport inventories still recommends Zurich; withdrawal of Singapore's historical school-stage travel proxy also preserves this recommendation. Evidence improves precision and changes decisions before it necessarily reduces the possible-champion set.

As a method diagnostic, TOPSIS and VIKOR use the same illustrative midpoint utilities and weights. VIKOR balances total and largest weighted shortfalls equally and uses sample best/worst values [4]. TOPSIS selects Zurich at the midpoint, whereas VIKOR selects Dublin; Kendall's $\tau_b = 0.483$. Dublin satisfies VIKOR's advantage and stability conditions. Under equal weights, Oslo has the lowest VIKOR index but fails those acceptance conditions, so it is not an accepted unique compromise solution. The midpoint comparison does not replace the interval decision in Equation 4. Agreement indicates similarity within the chosen conventions; disagreement exposes method dependence rather than justifying a post-hoc choice of the preferred winner.

6 Model revision and extension

6.1 Considering future changes

We examine three conditional changes while holding anchors and population denominators fixed. Resource expansion raises higher-education, stop and pool densities by 10%. Living pressure raises living costs, housing cost ratios and travel durations by 10%. Balanced improvement reduces pupil–teacher ratios, $PM_{2.5}$ and travel durations by 10%. A comparable multi-year panel is not available, so these changes have no estimated annual rate or forecast date.

For raw multiplier $r > 0$, the utility transformations are

$$T_r^+(u) = \frac{ru}{1-u+ru}, \qquad T_r^-(u) = \frac{u}{u+r(1-u)}. \tag{6}$$

These increasing maps propagate raw-scale changes through benefit and cost intervals. For known mean travel duration, the new Jensen interval is $[e^{-r\bar{t}/30}, 1]$. Singapore instead has bounds $[L_q, U_q]$ on expected utility. With $q = e^{-t/30}$ and $t' = rt$, Jensen and $0 \le q \le 1$ give

$$\mathbb{E}[q^r] \in \begin{cases} [L_q, U_q^r], & 0 < r < 1, \\ [L_q, U_q], & r = 1, \\ [L_q^r, U_q], & r > 1. \end{cases} \tag{7}$$

Raising both endpoints to r would generally misrepresent an expected utility. Equation 7 is an outer moment bound and need not be reversible. Fully missing travel remains unknown. Thus the extension accommodates the different information supplied by means and by travel distributions.

All three uniform-change scenarios retain Zurich as the conservative recommendation and all sixteen cities as possible champions under both weight references, including the Singapore-proxy withdrawal arm. Balanced improvement changes the fixed-weight second and third decisions from Singapore–Oslo to Oslo–Singapore. We also solve a decision boundary: holding other cities fixed, proportionally reduce one city's pupil–teacher ratio, $PM_{2.5}$ and travel durations until its maximum regret ties that of the best other city. Oslo reaches the tie at approximately 5.685% reduction, Singapore at 10.309%, and Stockholm at 28.875%. Zurich

already minimizes regret at baseline. A slightly greater reduction produces a strict recommendation switch in these calculations. These thresholds describe the model response to a specified joint change; they do not establish policy feasibility or a uniquely identified champion.

For a separate future-shock simulation, each draw combines a common shock and city-specific shocks under a declared bounded distribution, with the criterion weights held fixed. Original data intervals are not randomly completed. Specifically, $C \sim U[0,1]$ is shared across cities and $I_i \sim U[0,1]$ varies by city, independently; $\delta_i = 0.05C + 0.05I_i$. The relevant raw multipliers are $1 + \delta_i$ or $1 - \delta_i$, and affected criteria within a city share that multiplier. With 10,000 draws per scenario, Zurich's fixed-weight recommendation frequencies are 100.00%, 96.85%, 97.72% for resource expansion, living pressure and balanced improvement respectively. These are conditional stress frequencies under an uncalibrated distribution. Weight perturbations below answer a separate question and are not blended silently into the future-shock frequencies.

6.2 Considering different preferences and development situations

Education priority doubles the weights of the two education indicators. The independent living scenario doubles travel and transport-stop weights. Equation 1 then normalizes all weights. Education priority changes the conservative recommendation from Zurich to Oslo, with decision-rank $\tau_b = 0.700$ relative to baseline. Independent-living priority retains Zurich and gives $\tau_b = 0.917$. All sixteen cities remain possible champions in each profile. The resulting differences show how the definition of a desirable city affects the decision. They do not establish measured differences between fourteen- and eighteen-year-old students.

6.3 Perturbing weights

For each preference centre w', we draw $w^{(b)} \sim \text{Dirichlet}(\kappa w')$, with $\kappa \in \{20, 100, 500\}$ and 10,000 draws per setting. Every draw recomputes interval TOPSIS and maximum regret without filling missing values. We report mean decision rank, fractional first-place recommendation frequency and fractional top-three decision membership; ties share membership.

Table 3: Conditional recommendation frequencies (%) at concentration 100.

Preference centre	Zurich	Oslo	Singapore
Baseline	43.34	28.00	28.23
Education priority	6.83	45.62	42.11
Independent living	58.99	15.46	25.12

At concentration 100, Oslo has a better mean decision rank (2.1062) than Zurich (2.2305), although Zurich has the greater recommendation frequency and is the fixed-weight minimax choice. At concentration 20 under education priority, Singapore is the most frequently recommended city, whereas the fixed education-priority decision is Oslo. These two summaries optimize different quantities. We retain minimum maximum regret as the predeclared decision rule; Monte Carlo reports its sensitivity. Neither average ranks nor a network aggregation resolves the missing empirical information. A low AHP consistency ratio and repeated simulations demonstrate calculation properties, not representative student opinion.

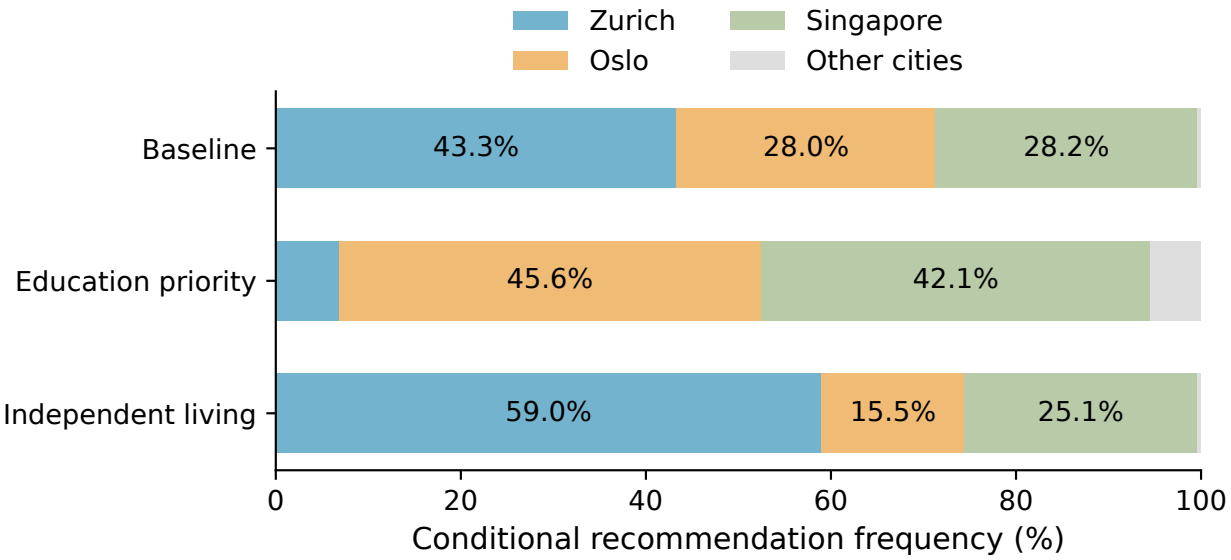


Figure 2: Recommendation frequencies under three declared preference centres, Dirichlet concentration 100 and 10,000 draws per centre. Frequencies are conditional on analyst-selected weight distributions, not probabilities of true superiority. Source: unchanged city evidence and declared preference multipliers.

6.4 Within-city fairness and growth stages

Citywide facilities per resident measure supply, not who can use it. Fees, opening hours, transport links, disability access and neighbourhood location can separate nominal supply from student opportunity. Without matched student locations and qualifying facilities, the city-level matrix cannot identify these access distributions.

A separate Shanghai school-resource case uses 16 districts, 235,542 regular senior-secondary pupils and 23,999 full-time teachers in the 2025 school year. The pooled utility is 0.604481; the student-weighted mean of district utilities is 0.603267. Ordering districts by resource utility and averaging the bottom 20% of pupil mass, with fractional inclusion of the boundary district, gives 0.580954, 0.022313 below the weighted mean. These are district-homogeneous school-location resource scenarios, not observed individual teacher exposure or resident accessibility [15]. To extend the same education criterion to districts d, let n_d be the matched pupil count and r_d its pupil–teacher ratio. The student-weighted utility is $\bar{u} = \sum_d n_d \, 15/(15 + r_d) / \sum_d n_d$. Its difference from a citywide ratio reflects spatial aggregation, not an effect of any particular policy. A school-location analysis also does not locate resident students or establish their travel access.

For future access measurement, define a_s as a student's qualifying access measure, incorporating the relevant eligibility, price and time conditions. With student weights p_s, report the weighted mean and the bottom-20% mean using fractional inclusion at the threshold. A common safety or travel standard could also be reported as weighted coverage. These require additional matched data; substituting a municipal average would not measure fairness.

A historical Istanbul household survey offers current-high-school groups aged 14–15 and 16–18. Their travel-utility marginal bounds overlap: $[0.35961, 0.55711]$ and $[0.29347, 0.59346]$, respectively. They include unknown durations and optional members; shared unknown-age mass means their extrema need not be jointly attainable. They describe 2012 broad groups, not exact ages fourteen and eighteen today. Singapore's MOE table records 38,555 students

aged fourteen and only 165 aged eighteen at the start of 2020 still in secondary schooling; the latter is not an all-eighteen-year-old resident denominator. These counts cannot be joined to census school-stage travel bins without a matching population/reference-day table [16, 17]. We therefore separate two mechanisms: age can change the observed indicators (travel, permissions or opportunities), and preferences can change the weights. An evidence-based age comparison would require $u_{ij}^{(14)}$ and $u_{ij}^{(18)}$ for comparable residents, then independently elicited $w_j^{(14)}$ and $w_j^{(18)}$ where available. The present preference scenarios test priorities without assigning unsupported age labels or pretending that all students face the same access.

7 Comparing cities within the same country

Every primary city belongs to a different country, so the main sample cannot identify within-country city contrasts. We use the existing United States and India cases as separately versioned examples with the same ten indicator definitions and scoring formula. They do not enter the sixteen-city ranking, and comparison within a country does not remove differences in source year, city boundary or educational coverage.

Table 4: Separate historical country cases using the legacy 25-city V117 evidence.

Country	City	Score interval
United States	Phoenix	31.39–71.68
	New York	30.22–72.81
India	Mumbai	28.82–60.64
	Delhi	32.59–63.90

Phoenix has higher retained affordability utilities than New York. Mumbai has the higher $PM_{2.5}$ utility, while Delhi has higher living-cost, housing, school-resource and broadband utilities in the retained evidence. The cases inherit the historical study's geographic and educational scope; their score pairs overlap, so neither case identifies a necessary overall winner. Overlapping score intervals leave both cities possible winners within each pair. Dimension-specific differences are more informative than attributing every score difference to a national effect. No causal national explanation is identified by these descriptive cases.

Reproducibility and integration note

The research archive preserves the unchanged team CSV, admitted-source hashes, utility bounds, ranking tables, scenario multipliers and simulation seeds. The frozen primary evidence version is V121; the current extensions are V122. Historical country and district/age cases retain their own versions and scope. Independent checks reproduce the score formula, competitor self-exclusion, recommendation frequencies and relevant transformation bounds. The source manifest records original files separately from derived extracts.

This document is a standalone chapter contribution, not a complete contest submission. The shared manuscript, summary sheet, news article, complete references and AI-use report require their own integration and final review.

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